

Fig. 2: Actual-size PCB pattern of the GSM based notice board

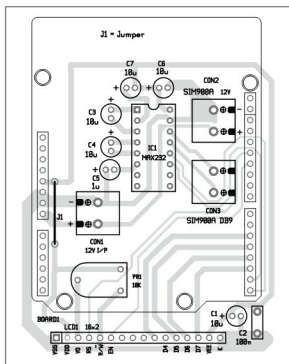


Fig. 3: Component layout of the PCB

levels are at $\pm 12V$, whereas both MCU input and output operate at 0V to +5V. Since RS232 is not compatible with the MCU, MAX232 (IC1) is utilised to enable the communication between the GSM modem and Arduino Uno by converting RS232-level signals to TTL-level signals. Rx (DB9.3) and Tx (DB9.2) pins of the GSM modem are connected through

IC1 at Tx and Rx pins of Arduino, respectively.

Software

The code written in Arduino can communicate with the GSM modem using AT commands. AT commands are sent or received from the modem using serial communication functions provided by Arduino library.

Serial.begin(9600) function helps to initialise the serial port with a baud rate of 9600.

Serial.available() function is always called before reading a data byte from the serial port of Arduino. This function returns a positive value when the data byte reception is complete; otherwise, it returns a non-positive value. It serves the purpose of waiting till the data byte reception is complete, so that there are no errors in reading the data byte.

Serial.read() function reads a data byte from the serial port of Arduino. It can return the data byte, which can then be stored in a variable or used for some condition check.

GSM modem responds OK when it receives command AT.

This is the best way to check communication between the modem and the MCU.

+CMGF command is used to set the SMS mode. Either text or PDU mode can be selected by assigning 1 or 0 in the command.

+CNMI AT command is used to

specify how newly-arrived SMSes should be handled. It can be used to set the GSM/GPRS modem or mobile phone either to route the received SMS messages directly to the serial port or to save these in message storage and then notify the host device by sending a string through the serial port.

For example, to route the received message string to the serial port of the device, the code written for this project uses AT + CNMI = 2,2,0,0,0 command.

Once an SMS is received, the GSM modem sends a long string to Arduino board, which consists of many details regarding the message including the original text. The code is written in such a way that it waits till any character is received through the serial port. Once it starts receiving, it checks for the occurrence of sixth ' ' and writes the rest of the characters directly into the 16x2 LCD. Characters after the sixth ' ' in the received string is the original text.

Construction and testing

An actual-size, single-side PCB pattern of the wireless notice board is shown in Fig. 2 and its component layout in Fig. 3. For convenience, we have designed the PCB as an Arduino shield. You can modify the design as per requirement. Also, you can test PCB with Arduino board using a cable connector.

Download the program in Arduino. Connect CON2 and CON3 with the GSM modem power and DB9 connector, respectively. Insert a SIM card in the GSM modem SIM slot. Power on the PCB using 12V supply to CON1. Send the SMS to the inserted SIM card number. LCD1 as well as Arduino serial port will display the message. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



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